

Hanyu Wang

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EDUCATION

- **Yale University | Master of Science** New Haven, CT, USA
Epidemiology of Infectious Diseases (Quantitative Specialization) *Expected May 2026*
Relevant Coursework: *Quantitative Methods for Infectious Disease (R), Transmission Dynamic Models for Infectious Diseases (R), Multivariate Statistics (R & SAS), Global Change and Infectious Disease*
- **University of Toronto | Honours Bachelor of Science** Toronto, Canada
Statistics Major, Quantitative Biology Major *Jun 2025*
Relevant Coursework: *Time Series Analysis (R), Methods for Multivariate Data (R), Machine Learning (R & Python), Experimental Analysis (R), Population Ecology (R), Computer Programming (Python)*

WORK EXPERIENCE

ENVIRONMENTAL INFECTIOUS DISEASES

- **Climate-sensitive Mosquito Population Dynamics and Disease Risk** Jan 2026-Present
Yale Center on Climate Change and Health, Yale University *Graduate Student Researcher*
 - Contributing to the Infectious Diseases section of the *Climate Change and Health in Connecticut 2025 Report*, focusing on mosquito population dynamics and mosquito-borne disease risk.
 - Analyzing **statewide mosquito trapping and arbovirus surveillance data** and **CDC disease surveillance data** to assess **spatial and temporal patterns** in mosquito abundance, species diversity, and arboviral infection rates.
 - Reporting analysis on **the effects of climate change** on mosquito population dynamics and mosquito-borne disease activity.
- **Meta-analysis of Land Use Impacts on Tick Populations and Tick-borne Disease** Nov 2025-Present
School of the Environment, Yale University *Co-author*
 - Co-authoring a systematic review and meta-analysis quantifying the impact of land use change on tick populations and tick-borne disease risk.
 - Conducting literature screening and data extraction, applied **quantitative meta-analysis methods**, supporting synthesis of heterogeneous study findings and manuscript preparation.
 - Identifying key drivers associated with disease risk, strengthening evidence for **environmentally mediated transmission pathways** and informing **tick-borne disease risk assessment frameworks**.

TRANSMISSION DYNAMIC MODELING

- **Stochastic SEIRS Modeling with Spatial-Ecological Dynamics** May–Aug 2024
Laboratory for Industrial and Applied Mathematics, York University *Research Assistant*
 - Optimized an **SEIRS model** to simulate infectious disease transmission with **demographic and biological factors**.
 - Developed simulations and computational pipelines, and performed **parameter estimation** and **sensitivity analysis**, incorporating **stochasticity and spatial heterogeneity dynamics**.
 - Enhanced model interpretability and realism, and identified **environmental mediators** as key intervention drivers, supporting policy-oriented disease modeling.
- **Multi-Factor Epidemic Modeling and Deep Learning Systems** May–Aug 2023
Center for Interdisciplinary Research in Mathematics and Life Health, Xi'an Jiaotong University *Research Assistant*
 - Investigated epidemic modeling involving **multi-factor co-evolution** for complex infectious disease systems.
 - Processed large-scale epidemiological data, implemented **computational models**, and developed hybrid deep learning-based modeling.
 - Strengthened capabilities in **complex system modeling** and data analysis, supporting development of **integrated, multi-dimensional disease risk models**.

TECHNICAL & RESEARCH SKILLS

Programming & Computational Tools

R (Advanced): statistical modeling, spatiotemporal analysis, data visualization

Python (Proficient): data processing, machine learning, stochastic simulation, scientific computing

Statistical & Mathematical Modeling

Transmission Modeling: structured compartmental models, mechanistic modeling, stochastic and spatially explicit simulations

Inference & Uncertainty: Bayesian hierarchical modeling, likelihood-based inference, parameter estimation, sensitivity analysis

Spatiotemporal Modeling: time-series analysis, spatial models, GLM/GAM, mixed-effects models

Research Domains & Applications

Infectious Disease Epidemiology: transmission dynamics, population-level inference, climate-sensitive disease systems

Disease Ecology: host–pathogen interactions, environmentally mediated transmission, multi-scale dynamics

Spatial Epidemiology: disease mapping, risk modeling, geospatial analysis of vector-borne diseases